FOUNDATION FOR FOOD & AGRICULTURE RESEARCH

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	*ROAR-0000000037*
Award Program	Rapid Outcomes from Agriculture Research (ROAR)
Project Title	*Development of a messenger RNA nanoparticle vaccine for prevention of African Swine Fever*
Grant Start Date	01/02/2022
Grant End Date	12/30/2022
Total Grant Budget	$290,000
Total FFAR Budget	$145,000
Matching Funder(s)	Genvax Technologies
Final Report Date	03/31/2023
Project Director/Principal Investigator Information	
Full Name	Dr. DL Hank Harris
Email	hank@genvax.com
Phone	515-450-0136
Grantee Organization Information	
Organization Name	Genvax Technologies, Inc.
Address	2503 S. Loop Drive, STE 5446
City, State Zip	Ames, IA 50010

Tax ID	86-2613060
Authorized Signing Official (ASO) Information	
Name	Mr. Joel Harris
ASO Title	CEO
Email	joel@genvax.com
Phone	319-530-2757

General Information

1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

Ames, Iowa, District 4

1.2. How many new jobs were created because of this grant funding?

None.

1.3. How many jobs were maintained because of this grant funding?

Three jobs were partially maintained by this grant.

1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

No.

1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

No.

2. Scientific Report

2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

An outbreak of African Swine Fever (ASF) disease in the United States would decimate the swine industry with severe global economic consequences (1) (6) (7). A 45% drop

in live hog prices is predicted in the first year of an outbreak in the U.S. (1). In 2020, there was once again an emergence of ASF within the western hemisphere, increasing the risk of spread to the U.S. Currently, there is only sporadic use of live attenuated vaccines in areas with endemic ASF. Their use complicates diagnostic testing, disease surveillance, and their long-term attenuation and safety has not been proven. A live attenuated ASF vaccine would not be suitable for protecting swine in areas that are currently ASF-free. Ideally, a safe and efficacious subunit vaccine would be ideal for protecting the U.S. swine industry from a potential ASF outbreak. To address this ongoing need, Genvax prepared a non-living ASF vaccine composed of messenger RNA combined with a nanoparticle. An experimental proof-of-concept study was performed in swine to determine if the vaccine would protect against a highly virulent form of ASF. The vaccine was prepared by Genvax Technologies Inc., an animal vaccine company located in Ames, Iowa. The proof-of-concept study in swine was conducted at the Plum Island Animal Disease Center, ARS, USDA. The vaccine elicited high levels of ASF specific antibodies in some of the vaccinated swine, but there was no delay in disease onset and no decrease in disease severity after viral challenge. Genvax will continue to work with Plum Island Animal Disease Center, ARS, USDA to gather additional insights from the data collected during the proof-of-concept study. Once a safe and efficacious non-living messenger RNA vaccine for ASF has been developed, Genvax plans to immediately seek approval from the USDA Center for Veterinary Biologics to deploy the vaccine in the United States should an outbreak of ASF occur.

2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

An experimental non-viable mRNA nanoparticle vaccine was developed for the prevention of ASF. In the proof-of-concept study the vaccine elicited high levels of ASF specific antibodies in some of the vaccinated swine, but there was no delay in disease onset and no decrease in disease severity after viral challenge. These data will provide insights for the development of improved subunit ASF vaccines. Genvax has now proven capable of manufacturing and delivering a highly multivalent mRNA nanoparticle vaccine coding for 22 different proteins within a defined timeline.

2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:

a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

Genvax prepared an experimental non-living ASF vaccine composed of messenger RNA combined with a nanoparticle that expressed 22 proteins of the ASF virus. An experimental proof-of-concept study was performed in swine to determine if the vaccine would protect against a highly virulent form of ASF. The vaccine elicited high levels of ASF specific antibodies in some of the vaccinated swine, but there was no delay in disease onset and no decrease in disease severity after viral challenge.

2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

Goal 1: Develop a messenger RNA/nanoparticle African Swine Fever vaccine.
● Objective 1: Use at least 22 structural and non-structural genes from the ASFV as transgenes in a self-amplifying messenger RNA vector to form an experimental vaccine in combination with a cationic nanoemulsion by July 2022.

Genvax successfully completed the goal of creating an experimental mRNA nanoparticle ASF vaccine. The vaccine contained mRNA coding for 22 different structural and non-structural genes from the ASF virus within a self-amplifying non-living vector. The RNA was then encapsulated within a lipid nanoparticle to form the final version of the vaccine. Genvax delivered the experimental vaccine in November 2022 to the Plum Island Animal Disease Center, ARS, USDA for an experimental challenge study in swine. The amount of vaccine delivered was enough for 10 pigs following a prime/boost/boost vaccination design. Each dose contained 25 ug of mRNA coding for each of the 22 different ASF transgenes.

Goal 2: Determine if the mRNA/nanoparticle vaccine will prevent ASF in swine.
● Objective 2: Determine if a combination of structural and non-structural genes derived from ASFV will induce immunity to ASF disease by November 2022.

Genvax successfully completed a proof-of-concept study in swine to determine if the experimental mRNA nanoparticle vaccine would protect against a highly virulent form of ASF in February 2023. The proof-of-concept study in swine was conducted at the Plum Island Animal Disease Center, ARS, USDA. 5 pigs were vaccinated with the experimental mRNA nanoparticle vaccine following a prime/boost/boost vaccination design. The vaccine elicited high levels of ASF specific antibodies in some of the vaccinated swine, but there was no delay in disease onset and no decrease in disease severity after viral challenge. Genvax will continue to work with Plum Island Animal Disease Center, ARS, USDA to gather additional insights from the data collected during the proof-of-concept study.

2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

Goal 1: Develop a messenger RNA/nanoparticle African Swine Fever vaccine.

● Objective 1: Use at least 22 structural and non-structural genes from the ASFV as transgenes in a self-amplifying messenger RNA vector to form an experimental vaccine in combination with a cationic nanoemulsion by July 2022.

During the development phase of Goal 1 we demonstrated that lipid nanoparticles delivered our self-amplifying messenger RNA vector more effectively than the proposed cationic nanoemulsions both in vitro and in vivo. Briefly, using commercially available Precision Nanosystems lipids we encapsulated our self-amplifying RNA vector in lipid nanoparticles using a NanoAssemblr Ignite+. Formulations were downstream processed using Amicon 10kDa MWCO centrifugal filters. Formulations were filtered through sterile 0.2 μm syringe filters inside a Biosafety Cabinet. Sucrose was added as a cryoprotectant and formulations were frozen and kept at -80 C. Formulations underwent testing for particle size, polydispersity index, total RNA concentration, and encapsulation efficiency. Dosing aliquots were shipped to PIADC for in vivo evaluation.

2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

Goal 1: Develop a messenger RNA/nanoparticle African Swine Fever vaccine.

● Objective 1: Use at least 22 structural and non-structural genes from the ASFV as transgenes in a self-amplifying messenger RNA vector to form an experimental vaccine in combination with a cationic nanoemulsion by July 2022.

Genvax successfully completed the goal of creating an experimental mRNA nanoparticle ASF vaccine. The vaccine contained mRNA coding for 22 different structural and non-structural genes from the ASF virus within a self-amplifying non-living vector. The RNA was then encapsulated within a lipid nanoparticle to form the final version of the vaccine. Genvax delivered the experimental vaccine in November 2022 to the Plum Island Animal Disease Center, ARS, USDA for an experimental challenge study in swine. The amount of vaccine delivered was enough for 10 pigs following a prime/boost/boost vaccination design. Each dose contained 25 ug of mRNA coding for each of the 22 different ASF transgenes. Genvax has now proven capable of manufacturing and delivering a highly multivalent mRNA nanoparticle vaccine coding for 22 different proteins within a defined timeline. Genvax vaccine production allows us to formulate vaccine from a sequenced pathogen within 3-4 weeks, making it an ideal process to deal with future swine and other livestock disease outbreaks.

Goal 2: Determine if the mRNA/nanoparticle vaccine will prevent ASF in swine.

● Objective 2: Determine if a combination of structural and non-structural genes derived from ASFV will induce immunity to ASF disease by November 2022.

Genvax successfully completed a proof-of-concept study in swine to determine if the experimental mRNA nanoparticle vaccine would protect against a highly virulent form of ASF in February 2023. The proof-of-concept study in swine was conducted at the Plum Island Animal Disease Center, ARS, USDA. 5 pigs were vaccinated with the experimental mRNA nanoparticle vaccine following a prime/boost/boost vaccination design. The vaccine was safe and did not cause any adverse effects after administration to the pigs. The vaccine elicited high levels of ASF specific antibodies in some of the vaccinated swine, but there was no delay in disease onset and no decrease in disease severity after viral challenge. Genvax will continue to work with Plum Island Animal Disease Center, ARS, USDA to gather additional insights from the data collected during the proof-of-concept study. These insights have the potential to inform future development of new ASF subunit vaccines.

2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

The only proven method to verify ASF vaccine efficacy is challenge studies performed in pigs as no in vitro data can predict parameters measured in pigs. Our collaborators at the Plum Island Animal Disease Center, ARS, USDA were responsible for the study design and protocol for the proof-of-concept study. Animal numbers were dictated by various factors including the upper limits of capacity of the animal facilities available. Due to inherently large biological variation between individual animals, there was not a sufficient number of animals in each experimental group to prove statistically significant differences. A successful proof-of-concept study result would be all treated pigs surviving without presenting any disease related clinical sign, whereas an unsuccessful test result would be animals showing mild to severe disease-related signs. The experimental challenge virus was intentionally selected to be consistently highly virulent and fatal as a stringent test of the novel countermeasures. These types of proof-of-concept studies are meant to guide future research directions and development of novel vaccines that can be further explored in statistically guided hypothesis testing studies.

2.8. List key stakeholders that could be served by results of this project.

Stakeholders impacted by an African Swine Fever (ASF) outbreak if there is no cure or vaccine include pork consumers, the pork industry, and supporting industries (1). Pork is the most widely eaten meat in the world (8). The U.S. exports 20% of its annual pork supply (9). Lack of U.S. pork in the world market would reduce protein availability and have negative consequences on food security (6) (7). The agriculture industry will feel widespread impacts if ASF enters the U.S. as movement of pigs would be halted, exports paused and pigs culled (1) Economic models estimate the worst case scenario of a U.S. ASF outbreak would result in a $50 billion loss to the US pig industry (1) (3). ASF can cause up to 100% mortality in pigs (1) and losing entire barns of pigs would decimate the income of U.S. producers and force layoffs, significantly reducing rural employment. The lack of production would decrease slaughter rates, causing downsizing in processing plants. Depopulation would reduce the need for feed, impacting corn and soybean markets (1). A U.S. outbreak of ASF would halt the export of pork and pork byproducts (internal organs, insulin, heart valves, etc.) (1). Without the ability to export, the U.S. market will flood with discounted pork, creating price deductions across substitute protein markets. "The availability of inexpensive pork in the U.S. domestic market would lead to price reductions in competing proteins such as chicken, eggs, and cheese," (1). As other countries adjust to the lack of U.S. pork, they will become accustomed to other protein sources, seriously impacting U.S. market share for years (1). The ripple effects will be felt across the US from the feed suppliers to the consumers. Currently, ASF is present in 23 countries in Africa, nine in Europe and over 10 in Asia (1). In 2018, the disease rapidly spread through China and some Asian countries previously unaffected by the virus despite the culling of millions of pigs (1) (6). It was estimated that in 2019, China lost over one third of its swine population due to ASF (2). In July 2021, ASF was confirmed in the Dominican Republic, the first time in over 40 years that the disease was confirmed in Central America (4). In September 2021, the disease was also confirmed in Haiti (5). Upon the one year anniversary of ASF confirmation in the wild boar population in eastern Germany, veterinary experts project that they will be dealing with ASF for the next five years (10). The recent introduction of Porcine Epidemic Diarrhea virus from China shows the vulnerabilities to the U.S. of transboundary and foreign animal diseases. The U.S. has an immediate need for a safe, efficacious and readily available cure or vaccine for ASF. An attenuated or modified live vaccine would not be ideal or recommended in the face of an outbreak in the United States. Genvax will continue to work with Plum Island Animal Disease Center, ARS, USDA to gather additional insights from the data collected during the proof-of-concept study. These insights have the potential to inform future development of new ASF subunit vaccines.

2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

Nothing to report.

2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

Nothing to report.

2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

Nothing to report.

2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

One undergraduate student intern was involved in this project during summer 2022. The student intern helped make plasmid DNA, perform fingerprint analysis, and gene cloning for a portion of the 22 ASF genes.

3. Information Products

3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

At this time, no information products have been produced from this project.

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3.2. Please provide a list of citations for the information products produced during the project.

At this time, no information products have been produced from this project.

3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

At this time, no information products have been produced from this project.

3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

At this time, no information products have been produced from this project.

3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

At this time, no inventions, patent applications, and/or licenses have resulted from this project.

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#	#	
Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	

3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

Genvax and USDA PIADC have entered into a confidentiality agreement. Depending on the results of the animal studies, a decision will be made by the two parties about the dissemination of results. Please see the corresponding Data Transfer Agreement (DTA_for_PIADC.pdf) and Non-Assistance Cooperative Agreement (NACA_Original 58-8064-1-009 - FE1.pdf).

3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?

Genvax is using an Electronic Notebook system (SciNote) to track all steps in the creation of our various inputs in the production process from initial cloning, sequence confirmation, plasmid production, in vitro transcription, down-stream purification, encapsulation steps. This data will be made available at the appropriate time when agreed upon by both Genvax and USDA PIADC.

4. Data Management

4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries). Please list the data generated for the award.

We generated data on the quantity and quality of our self-amplifying RNA vectors containing the 22 ASFV transgenes. RNA concentrations were analyzed with a NanoDrop One and found to be within specifications of OD260/280 ratios of at least 1.9 or higher for all saRNAs. A minimum yield of 1.5 mg saRNA was used for downstream formulations. Capillary Electrophoresis of the saRNA to determine intactness was performed for all 22 saRNAs and found to be the correct sized product and within specifications. After encapsulation of the saRNA within lipid nanoparticles we evaluated the encapsulation efficiency, encapsulated saRNA concentration, polydispersity index, and particle size. We have not yet received any data or statistical analysis for the in vivo challenge study. Genvax will continue to work with Plum Island Animal Disease Center, ARS, USDA to gather additional insights from the data collected during the proof-of-concept study.

4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

For each of the self-amplifying RNA vectors containing 1 of 22 different ASFV transgenes, we generated data on the vector and data on the lipid nanoparticle encapsulated formulation.

4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

Please see corresponding data files (Data Management 4.3).

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Dr. DL Hank Harris

PI Signature: DL Hank Harris Date: 03 / 29 / 2023

Authorized Signing Official (ASO) Name: Joel Harris

ASO Signature: Date: 03 / 29 / 2023

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